

SHOPSY E-COMMERCE DATABASE MANAGEMENT SYSTEM

PRODUCT AND CATEGORY MANAGEMENT SYSTEM

1. Database Design :

The ShopSphere Online E-Commerce Database Management System uses Product and Category as important entities for managing products available on the e-commerce platform. The Category table stores information about different product categories, while the Product table stores product details such as product name, price, brand, stock quantity, and product description. The Product table is connected to the Category table using Category_ID, which establishes a one-to-many relationship between Category and Product.

1.1 Category Table :

Attribute	Key Type
Category_ID	Primary Key
Category_Name	Unique Key
Description	-

1.2 Product Table :

Attribute	Key Type
Product_ID	Primary Key
Product_Name	Not Null
Seller_ID	Foreign Key
Category_ID	Foreign Key

Attribute	Key Type
Price	-
Stock	-
Description	-

2. Primary and Foreign Key Relationships :

The relationship between Category and Product is defined using Category_ID.

- Category.Category_ID → Primary Key
- Product.Product_ID → Primary Key
- Product.Seller_ID → Foreign Key
- Product.Category_ID → Foreign Key referencing Category.Category_ID
- Relationship: One-to-Many (1:M)
- One category can contain many products, while each product belongs to a category.

3. Store Product Details and Table Creation :

The following SQL statements create the Category and Product tables and store product details such as product name, seller, category, price, stock, and description. The Category table stores category information, while the Product table stores the details of products available in the ShopSphere e-commerce system. The relationship between Category and Product is maintained using Category_ID, while Seller_ID is used as a foreign key in the Product table.

3.1 Create Category Table :

```
SQL> CREATE TABLE Category (  
2     Category_ID NUMBER PRIMARY KEY,  
3     Category_Name VARCHAR2(100) UNIQUE,  
4     Description VARCHAR2(200)  
5 );  
  
Table created.
```

3.2 Create Product Table :

```
SQL> CREATE TABLE Product (  
2     Product_ID NUMBER PRIMARY KEY,  
3     Product_Name VARCHAR2(100) NOT NULL,  
4     Seller_ID NUMBER,  
5     Category_ID NUMBER,  
6     Price NUMBER(10,2),  
7     Stock NUMBER,  
8     Description VARCHAR2(500),  
9     FOREIGN KEY (Category_ID) REFERENCES Category(Category_ID)  
10 );  
  
Table created.
```

4. Product Insertion, Updating and Deletion :

4.1 Category Records :

```
SQL> INSERT INTO Category VALUES (1, 'Electronics', 'Electronic products');  
1 row created.  
  
SQL> INSERT INTO Category VALUES (2, 'Fashion', 'Fashion and clothing products');  
1 row created.  
  
SQL> INSERT INTO Category VALUES (3, 'Home Appliances', 'Home appliance products');  
1 row created.  
  
SQL> INSERT INTO Category VALUES (4, 'Sports', 'Sports and fitness products');  
1 row created.  
  
SQL> INSERT INTO Category VALUES (5, 'Books', 'Books and educational products');  
1 row created.
```

4.2 Product Records :

```
SQL> INSERT INTO Product VALUES (101, 'Wireless Headphones', 1, 1, 1499.00, 25, 'Bluetooth wireless headphones');
1 row created.

SQL> INSERT INTO Product VALUES (102, 'Cotton T-Shirt', 2, 2, 599.00, 40, 'Comfortable cotton t-shirt');
1 row created.

SQL> INSERT INTO Product VALUES (103, 'Electric Kettle', 3, 3, 1299.00, 20, 'Stainless steel electric kettle');
1 row created.

SQL> INSERT INTO Product VALUES (104, 'Running Shoes', 4, 4, 1999.00, 15, 'Lightweight running shoes');
1 row created.

SQL> INSERT INTO Product VALUES (105, 'DBMS Fundamentals', 5, 5, 450.00, 30, 'Database management system book');
1 row created.
```

5.Data Retrival :

5.1 Display Category Records :

```
SQL> SELECT * FROM Category;
```

```
CATEGORY_ID
```

```
CATEGORY_NAME
```

```
DESCRIPTION
```

```
1
Electronics
Electronic products
```

```
2
Fashion
Fashion and clothing products
```

```
CATEGORY_ID
```

```
CATEGORY_NAME
```

```
DESCRIPTION
```

```
3
Home Appliances
Home appliance products
```

```
4
Sports
```

```
CATEGORY_ID
```

```
CATEGORY_NAME
```

```
DESCRIPTION
```

```
Sports and fitness products
```

```
5
Books
Books and educational products
```

5.2 Display Product Records :

```
SQL> SELECT * FROM Product;
```

```
PRODUCT_ID
-----
PRODUCT_NAME
-----
SELLER_ID CATEGORY_ID PRICE STOCK
-----
DESCRIPTION
-----
101
Wireless Headphones
1 1 1499 25
Bluetooth wireless headphones

PRODUCT_ID
-----
PRODUCT_NAME
-----
SELLER_ID CATEGORY_ID PRICE STOCK
-----
DESCRIPTION
-----
102
Cotton T-Shirt
2 2 599 40
Comfortable cotton t-shirt

PRODUCT_ID
-----
PRODUCT_NAME
-----
SELLER_ID CATEGORY_ID PRICE STOCK
-----
DESCRIPTION
-----
103
Electric Kettle
3 3 1299 20
```

6. Category Update and Delete :

```
SQL> UPDATE Category
2 SET Category_Name = 'Mobile Accessories'
3 WHERE Category_ID = 1;
```

```
1 row updated.
```

```
SQL> DELETE FROM Category
2 WHERE Category_ID = 5;
```

```
1 row deleted.
```

7. Product Update and Delete :

```
SQL> UPDATE Product  
  2  SET Price = 1699.00  
  3  WHERE Product_ID = 101;
```

```
1 row updated.
```

```
SQL> DELETE FROM Product  
  2  WHERE Product_ID = 102;
```

```
1 row deleted.
```